SPE-227396-MS

Successful Development and Application of a Compression-Set Packer for Casing Cementing

Shobeir Pirayeh Gar, Halliburton, Carrollton, Texas, US; Allan Zhong, Halliburton, Tuas, Singapore; Reena Thomas and Priyansh Desai, Halliburton, Houston, Texas, US

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This paper presents the design development, validation testing, and field application of a compression-set packer. It has been successfully installed in multiple-stage cementing operation for both 9 5/8-inch and 13 3/8-inch casing designs. The developed design ensures well integrity by integrating a packer and cementer into a single assembly and achieves proper zonal isolation. Additionally, the paper discusses the set-based design method and compares predicted performance to test results.

A set-based design approach was used along with engineering simulation, and the final element design was test-validated as Grade V3 per ISO 14310 and API 11D1. This is the most stringent rating criterion in a liquid medium requiring the tool pressure rating to be held for 15 minutes with less than 1% pressure drop. The element was first heated to 200 Fahrenheit, then set to seal against casing ID, followed by a sequential differential pressure of 6,000 psi from below and above. Post-test inspection was conducted to assess the packer deployment, and the test results were compared against simulation predictions.

The newly developed compression-set packers, designed using a set-based approach, were test-validated successfully, withstanding the required pressure rating of 6,000 psi at 200 Fahrenheit, and providing V3-rating isolation assurance. Post-test inspection confirmed that the packer deployed as predicted by engineering simulations. Field application has been highly successful, with 65 cementing jobs completed successfully in the Saudi field to date. The developed compression set packers significantly outperform traditional inflatable packers, which often exhibits deficiencies in withstanding moderate casing pressures (< 5000 psi), and act as a reliable mechanical barrier to ensure proper zonal isolation in casing-to-casing applications, mitigate sustained casing pressure (SCP), and preserve wellbore integrity.

The development process of compression-set packers, as one of the most recent successful well barrier tools for cementing applications, with a critical review of the mechanics, the design method, and the test results, is the novel contribution of this paper to the existing literature.

Introduction

Inflatable packers have been commonly used in most wells to seal the annulus, mainly to support second stage cementing pressures or short-term overpressures from below. However, they do not often provide a reliable barrier against long-term pressures and temperature cycles experienced during multi-stage cementing and may allow some pressure penetration. This may disturb the cement setting process, create sustained casing pressure (SCP), and compromise well integrity, which can result in loss of production and a costly repair plan (Halliburton 2021)

To enhance the reliability of the annular sealing and promote well integrity, a mechanically set packer can be designed to provide a premium seal by meeting an appropriate industry-standard-rated isolation assurance. The mechanically set packers typically refer to packers set by a compression force applied through a setting sleeve. These packers are usually classified as either gas-tight (V0-rated) or liquid-tight (V3-rated), depending on their application as per ISO 14310 (2008) and API 19AC (2016). To manage the cost of such a premium mechanical barrier compared to inflatable packers, the compression-set packer can be integrated with a cementer into a single assembly with a simple and reliable design (Halliburton 2021). In other challenging applications, compression-set packers have been shown in the past to be successfully integrated into one tool string for wellbore cleanout purposes (Maher et al. 2021).

Since compression-set packers achieve the robust state of sealing when they are fully squeezed, the assembly needs to accommodate an axial displacement (stroke) during the hydromechanical setting process and lock the stroke when the setting force is removed. Moreover, the integrated tool needs to have several dedicated cementing ports distributed along the cementer for a more uniform cementing operation. Hence, the overall length of the packer element becomes a critical design parameter affecting the seal's mechanical performance, the design geometry's compactness, and the drillability of the tool. Packer elements that are too short may not generate the required contact stress at the sealing surfaces when they are squeezed. On the other hand, packer elements that are too long may extrude over the backup shoes during the setting process. Therefore, there is an optimal value for the element length that needs to be determined via optimization analysis. Another critical design parameter is the casing inner diameter (ID), which is not a fixed number as typically deemed in analyses or tests, and indeed, it has variations in the real well that must be considered when determining the optimal length of the packer.

Extensive experience in packer analysis has shown that the optimal packer length and casing ID are strongly coupled due to their nonlinear interaction. This is because the casing ID acts as a nonlinear boundary condition for the packer element during setting. Thus, a reliable mechanically set packer should have an optimum length and be robust against casing ID variations for casing-to-casing applications. Finding the optimum packer length using traditional physics-based simulations is challenging, as running the simulation for all feasible combinations of packer length and casing ID is very costly and time-consuming.

One way to achieve this goal efficiently is through Set-Based Design (SBD), where a manageable number of design options or a set of possibilities are analyzed simultaneously to explore the design geometry space via Design of Experiment (DOE) and perform a regression analysis to form a surrogate model, which correlates the performance of the packer to its main design parameters. The performance of the packer can be defined as a multi-objective function of critical response parameters. An optimization analysis is then performed to find the best design from the surrogate model (Pirayeh Gar et al. 2024).

In this paper, the design development of a compression-set packer is first presented with a focus on the critical design considerations affecting the performance and functionality of the tool. A set-based design methodology is then presented, followed by a series of Finite Element Analyses (FEA) to conduct a DOE study, where the best compression-set packer design can be identified. The authors highlight the important analytical considerations for realistic simulation of mechanically set packers. In the next step, the validation testing is presented to discuss the test setup and conditions, as well as experimental observations. The test results are then compared against the analytical predictions to verify the design concept. The field application and performance of the newly developed compression-set packer is briefly reviewed, including the tool's rate of success as well as any reported operational concerns. The authors will conclude their work by providing a summary of what was accomplished, specific findings, lessons learned, future work, and room for improvement.

Design Development of a Compression Set Packer

The developed compression-set packers include elastomeric elements that are hydromechanically set using a setting sleeve pushed by a seated plug pressurized via internal casing pressure. The packer contracts axially under the setting load and expands radially to seal the annulus. The inner diameter of the packer (ID) seals against the mandrel outer diameter (OD), and the packer OD seals against the casing ID. An internal continuous ratcheting mechanism is employed via a body lock ring component to keep the squeezed packer locked in place and lock the stroke once the packer is set. At this point, the opening sleeve is further pushed by an internal casing pressure to shear a dedicated drive-pin system and expose the cementing ports for the second stage of cementing. The drive-pin system is designed to eliminate the risk of premature rupture-disc failure often seen in traditional designs, thus preventing any potential leak path from the setting operation. The cementer is equipped with six dedicated cement ports to provide an improved and uniformly distributed 360° cement flow with the flow rate capacity of about 16 barrel per minute (bpm).

Fig.1 illustrates the overall layout of the cementer with the mechanically set packer. The packer has two rubber elements with a metal spacer in between and two metallic support shoes at both ends. The backup shoes plastically deform and deploy concurrently with the packer elements to form a robust seal and avoid any extrusion of the rubber element and loss of seal integrity. The compact design of the integrated stage cementer and the annulus-sealing packer results in a significant reduction in drillable material. The packer is designed to offer a bi-directional seal under 6,000 psi pressure and at 200 degrees Fahrenheit temperature.

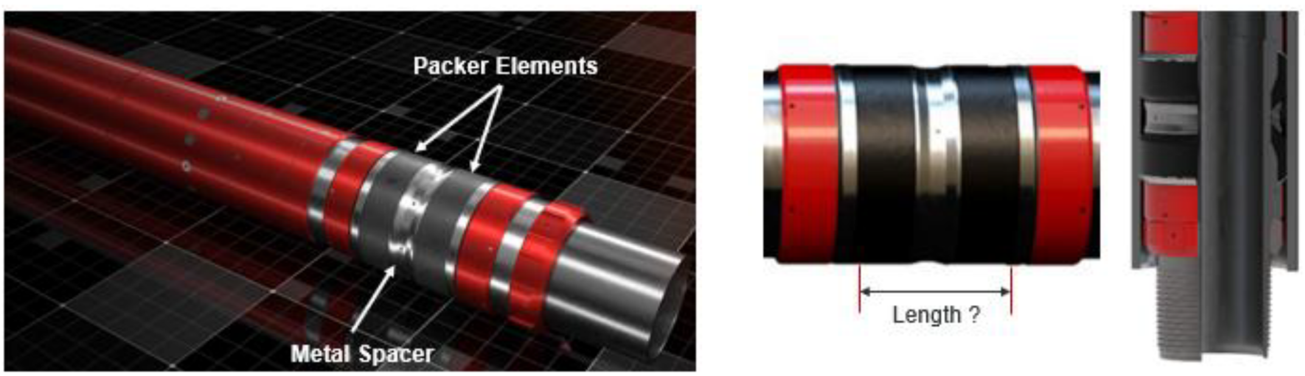


Figure 1—Overall layout of the compression-set packer

The design development process starts off by establishing a good base design using principles of engineering mechanics, domain knowledge, and operational considerations. The base design offers a working solution, but not necessarily the best, where the focus is on the overall setting performance in terms of the required setting load, robustness of the backup shoes, and interaction between packer elements and the metal spacer. The body lock ring is designed to accommodate the estimated stroke during the set with minimum backlash to minimize the loss of setting force.

Once a base design is established, it needs to be optimized for a well-defined performance objective function. We are basically looking for the best packer length considering the casing ID variations. To simplify the optimization process and the number of the design parameters, the robustness of the backup shoes and other deployable components are studied and improved before the main optimization analysis. As discussed in the next section, an SBD approach is taken to explore the design space for the best packer length vs. different casing IDs using the DOE technique. The design optimization analysis is followed by a comparative FEA and validation testing before preparing for field application.

Optimization Analysis

Complex systems like packer assemblies often exhibit highly nonlinear responses and interactions between their components, making traditional physics-based optimization analyses computationally costly and resource-intensive. Therefore, the combined data-driven and physics-based approaches are employed for comprehensive and fast exploration of the design space.

Fig.2 illustrates the physics-based model of the packer and the overall optimization analysis workflow. As mentioned earlier, the main design parameters of the compression-set packer assembly for optimization analysis are chosen as the packer length and casing ID. A design space is defined based on a feasible range of the design parameters. The design cases are sampled using the Latin Hypercube Sampling (LHS) method to minimize the overlap between the samples. Finite element analysis is conducted on each design sample to study the performance of the packer. This is called a Design-of-Experiment (DOE) approach. A multilayer neural network (NN) is then used to take the simulation results as training data and generate a surrogate model as the data-driven model. The surrogate model correlates the design parameters to a performance objective index (POI), which is a multi-objective function of the main response parameters. The optimization analysis is conducted using a genetic algorithm to find the optimum design from the surrogate, where POI reaches its global optimum. At the end, the optimum design predicted by the surrogate model is compared with finite element analysis results to evaluate the surrogate approximation. An acceptable proximity of ± 5% is applied when comparing the surrogate predictions with FEA results. If the difference is larger than 5%, more samples need to be added to the design space to refine the surrogate model.

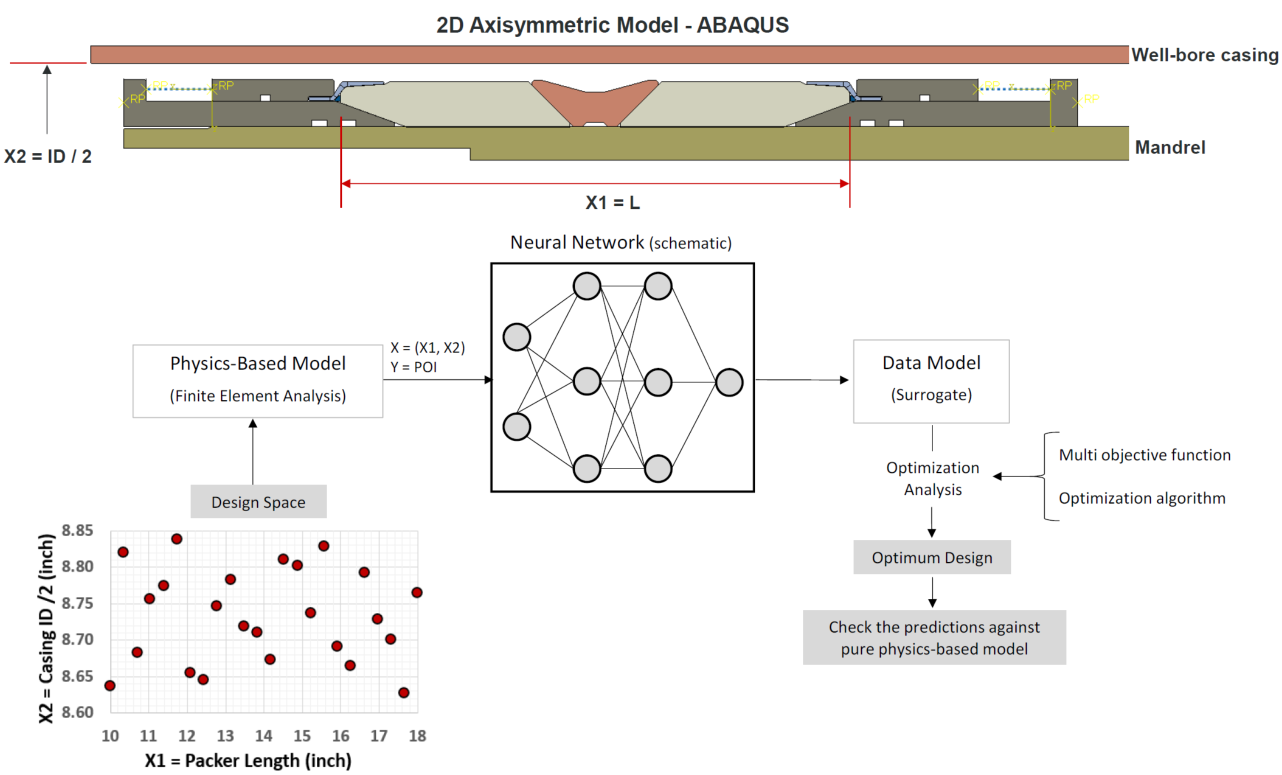
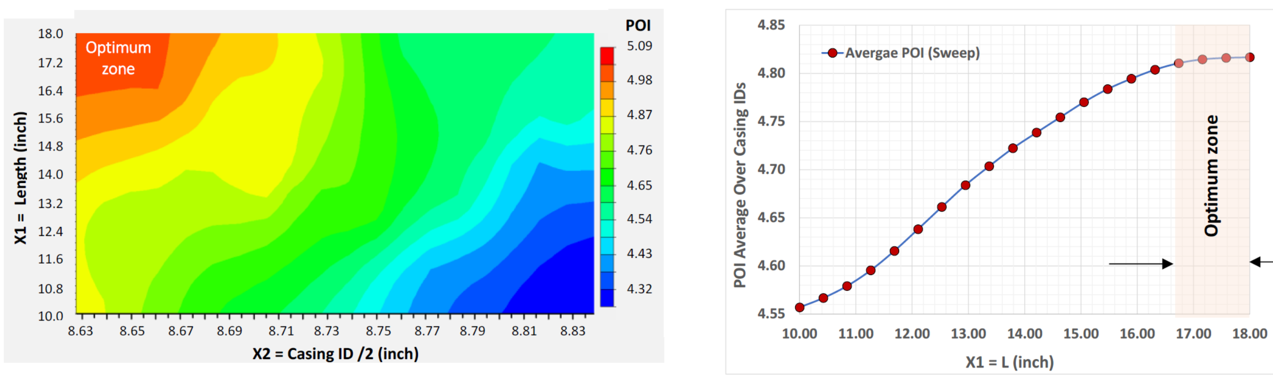


Figure 2—Optimization analysis workflow

Twenty-four design samples, possessing different lengths and casing IDs, were analyzed through finite element simulation to find the performance objective index for each design. POI was defined as a normalized function of the required setting force, the contact stress at the casing ID and at the mandrel OD, and the amount of force that can be transferred from the setting side to the opposite side of the packer. It is desirable for a packer to have lower required setting force, higher contact stress at the sealing surface, and higher force transfer coefficient. A lower setting force simplifies the design of the setting sleeve as well as the setting operation, a higher contract stress at the sealing surface enhances the seal robustness, and a higher force transfer coefficient ensures the packer sets uniformly from both ends. Past project experience and domain knowledge are used to normalize the main response parameters against a benchmark.

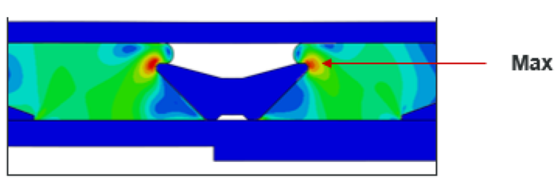
Fig. 3 presents the 2D contour and 1D graph representations of the surrogate model. As seen in the 2D contour, the surrogate model correlates the POI with the design parameters — packer length (X1) and casing ID (X2) — and implies that the optimal design packer length is between 16 to 18 inches. To further refine the optimum zone, the POI for any specific packer length was averaged over the entire casing ID domain to represent a 1D graph. Averaging POI over the range of casing ID (sweeping) is a very practical consideration to account for casing variation effects on the optimum packer element length. With further refinement, it was observed that the best packer length lies within the 17 to 18 inches range. To pick the most robust solution, both packer lengths of 17 and 18 inches were selected for further studies via FEA, including the differential pressure of 6,000 psi.



(a) (b)

Figure 3—Surrogate model of the compression-set packer: (a) 2D contour representation of POI; (b) 1D graph representation of POI

Fig.4 shows the principal strain contour of the rubber element with clear concentration at the vicinity of the metal spacer under 6,000 psi pressure. The magnitude of the rubber strain for the 17-inch-long and 18-inch-long packers were respectively found as 96% and 102%. Thus, the 17-inch-long packer with the smaller rubber strain was selected as the optimal design. To validate the accuracy of the surrogate model, FEA was done on the optimal design for both Max and Min casing IDs to compare the POI.



L = 18 inch ---- Max rubber strain =102 % (> 100%)

L = 17 inch ---- Max rubber strain = 96 % (< 100%)

Figure 4—Location of the maximum principal strain of the rubber under 6,000 psi pressure

Table 1 presents the comparison between FEA and surrogate predictions and shows the difference of about 2% (< 5%), which is deemed as an acceptable analytical proximity. The results showed that the surrogate model was refined enough to reasonably cover the design space. It is also noted that the POI for Min Casing ID was somewhat larger than that for Max casing ID, implying that the design was governed by the Max casing ID scenario, as it is more challenging to achieve the full seal in Max casing ID compared to Min casing ID. This is well in line with the principles of mechanics and our past experiences. The optimum packer length is now confirmed to proceed with validation testing in the next step.

Table 1—Comparison between POI prediction of FEA and the surrogate model

Design Case	Surrogate Prediction	FEA Prediction	Difference (%)
L = 17 inches with Max Casing ID	4.57	4.48	2.04
L = 17 inches with Min Casing ID	5.07	4.97	2.06

Validation Test

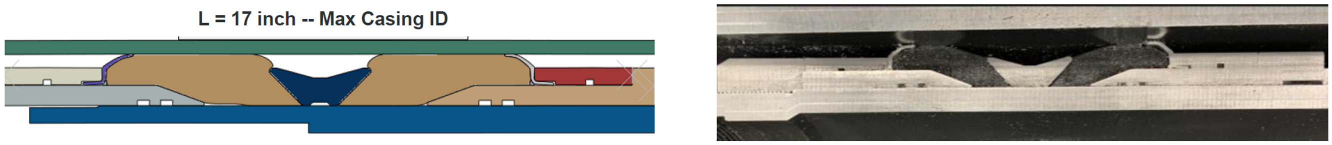
The optimum packer design was manufactured and prepared for Grade V3 validation testing per ISO 14310 (2008) and API 19AC (2016). The Grade V3 is the most stringent liquid rating that requires the packer to hold the target pressure for 15 minutes with less than 1% pressure drop. This ensures that the packer has passed a liquid test, along with the requirements for differential pressures and temperature cycles. As mentioned earlier, the compression-set packer was designed for 6,000 psi pressure under 250°F temperature.

Fig. 5 shows the test setup in a closed casing with hydraulic pressure systems to incrementally apply pressure using UCON as the testing fluid. The pressure was read from the hydraulic pump, and the stroke length of the packer was measured using Linear Variable Displacement Transducers (LVDTs). The shear-pin failure was identified by a sudden shift in the sleeve position. The 17-inch-long compression-set packer passed the Grade V3 validation test robustly without exhibiting any signs of a leakage or local damage to the rubber elements and backup systems. The packer assembly and the casing were bisected for post-test inspections and for comparing the observations with FEA predictions as discussed in the next section.

Figure 5—Validation test setup

Comparative Analysis

Fig. 6 shows the comparison of the packer setting performance at Max casing ID between the FEA prediction and validation test inspection. The packer assembly was bisected for post-test inspections and somewhat relaxed due to the release of the tool. The overall packer set configuration, particularly the deployed shape of the backup shoes, the required setting force, and the total stroke length were all found in good agreement with FEA results. No major rubber damage or extrusion was observed at the vicinity of the metal spacer, as predicted by the analysis.



(a) (b)

Figure 6—Packer setting comparison at Max casing ID: (a) FEA; (b) validation test

Field Application

The compression-set packer discussed in this paper was developed for 9 5/8-inch and 13 3/8-inch sizes, and it has been deployed in a Saudi field over 132 times to date with 100% success rate of deployment. In some irregular scenarios, however, the sliding components could not open the second stage cementing port under the target pressure level. This is currently under investigation and improvement. One of the key design considerations is to have a robust manufacturing tolerance particularly for tight gaps to make sure the sliding sleeves are not caught or notched by any sharp edges when opening/closing the cement port. Also, to open/close the cementing port at the target pressure, it is critical to minimize the variations in the shear strength of the drive-pins and to make sure they fail under pure shear mechanism and not a combined flexure-shear mechanism, which typically requires a higher shear load. Simplifying the design by reducing sliding components and minimizing the usage of aluminum material for enhanced drillability are some of the key future design considerations for future improvements.

Conclusions

A set-based design (SBD) methodology was adopted to develop a compression-set packer for casing cementing. The packer was designed and optimized for the best setting performance using the physics-based and data-driven approaches via the Design-of-Experiment (DOE) technique. The Grade V3 validation testing was found successful and in line with FEA results. The field applications of the tool have been extremely successful with over 132 times packer deployment with different casing ID variations. Some more specific findings and considerations can be summarized as:

1. To develop a robust design, it is critical to first establish a good base design using principles of engineering mechanics and domain knowledge. A sizing optimization analysis can then be conducted to find the most robust packer dimensions.
2. Packer length and casing ID were found as the two critical design parameters with highly non-linear interactions as suggested by the surrogate model. It is recommended to find the optimal packer length by averaging over the range of casing IDs (sweeping method).
3. Combining physics-based and data-driven models to build a predictive surrogate was found very effective in enabling the fast-track exploration of the design space. The governing design parameters, definition of the performance objective index (POI), and proper proximity between FEA and surrogate model predictions (within ±5%) are the key analysis considerations. The recommended response parameters include setting force, contact stresses at sealing surfaces, rubber principal strain, backup shoe plastic strain, and the percentage of the setting load that is transferred to the other side of the packer.
4. The robust performance of the packer was indicated by the full deployment of the backup shoes on both sides which occurred concurrently with the packer element deployment. This reduced the risk of element extrusion over the backup shoe, rubber nibbling or local damage, as well as unsymmetric deployment of the backup shoe.

5. The developed compression-set packer achieved a robust seal and benefitted from an enhanced squeeze using the prop element design. The metal spacer between the elements helped to control the element setting and reduce the risk of element extrusion over the backup shoes.
6. The future work may include further simplification of the design to have a smaller number of sliding components, enhancing the drill-ability by replacing aluminum sleeves with composite sleeves, improving the design for tight tolerances, and a scalable design of the drive-pin system.

Acknowledgment

The authors would like to thank Frank Acosta, Halliburton's global technical advisor, for his technical insight and guidance during this project. They would also like to thank the management of Halliburton for their encouragement and permission to publish this paper.

References

API Spec 19AC. 2016. Specification for Completion Accessories.

Halliburton. 2021. https://www.halliburton.com/en/products/obex-isolock-packer-collar.

ISO 14310. 2008. Petroleum and natural gas industries — Downhole equipment — Packers and bridge plugs.

Maher, P., Nelson, C., and Dockweiler, D. 2021. Development and Implementation of Two Compression Set Tools in a Wellbore Clean Out String. *SPE Middle East Oil & Gas Show and Conference*, SPE-204664-MS. https://doi.org/10.2118/204664-MS.

Pirayeh Gar, S., Zhong, A., and Jacob, J. 2024. Machine Learning Aided Optimization of Non-Metallic Seals in Downhole Tools. *Engineering Modelling, Analysis and Simulation (EMAS) Journal*, Vol. **1**, Issue 1.